Multi-State Models for Event History Analysis

Based on Andersen PK & Keiding N (2002)

Ziye Lin - STAT 935

Toward Multi-State Models

Motivation:
Suppose we want to analyze a process that involve a sequence of events to study their occurrence times as well as the types of event that occur.

- Example: Tracking an individual's health pattern over time. The individual gets sick, recovers, gets sick, ..., and dies at one point.

This is an example of **event history analysis**. Frequently encounters in medical research, reliability engineering, economics and finance, and sociology and demography.

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Approach $\longrightarrow$ **Multi-State Models**

Multi-State Models

A **multi-state model** characterizes a stochastic process

$$(X(t), t \in \mathcal{T}),\ \mathcal{T} = [0, \tau)\ , \tau \leq +\infty$$

which consists of the following components:

1. a finite state space: $\mathcal{S} = \{1, 2, \dots, p\}$.
2. an initial distribution: $\pi_b(0) = \mathbb{P}(X(0) = b)\ ,\ b \in \mathcal{S}$.
3. transition probabilities:

$$P_{bj}(s,t) = \mathbb{P}(X(t) = j | X(s) = b\ ,\ \mathcal{X}_{s^-})\ ,\ j, b \in \mathcal{S}\ ;\ s, t \in \mathcal{T}\ ;\ s \leq t\ ,$$

   where $\mathcal{X}_t$ is a history (σ-field) generated by $X(t)$.

4. transition intensities:

$$\alpha_{bj}(t) = \lim_{\Delta t \to 0} \frac{P_{bj}(t, t + \Delta t)}{\Delta t}\ ,\ j, b \in \mathcal{S}\ .$$

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Each subject's event history is a realization of a stochastic process.

Multi-State Models

The two-state model:
The traditional survival analysis can be considered as the analysis of a multi-state process with two states: $\mathcal{S} = \{0, 1\}$

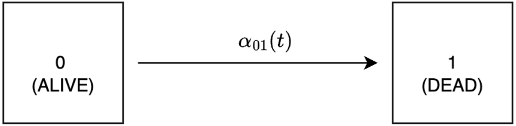

Figure: The two-state model for survival data

Note:

- $\alpha_{10}(t) = 0$ for all t as we cannot proceed from "DEAD" to "ALIVE"
- State 1 is called an **absorbing state** from which no further transition can occur, while State 0 is called a **transient state** as it is not absorbing.

The competing risks model

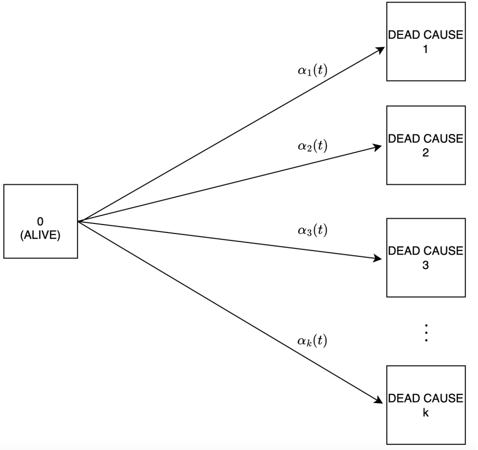

Figure: Competing risks model for mortality from k causes

The illness-death model

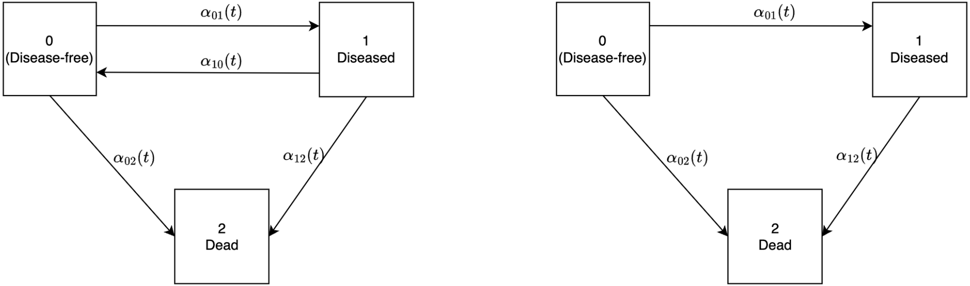

Figure: The illness-death model with reversibility (left) and unidirectional ilness-death model (right)

Recurrent events

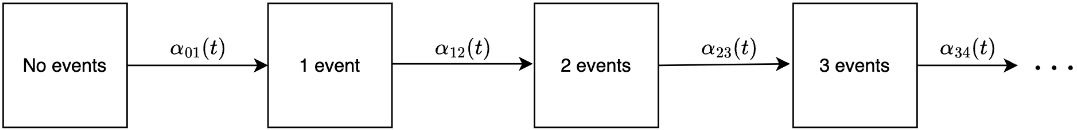

Figure: A model for recurrent events

Interaction between life history events

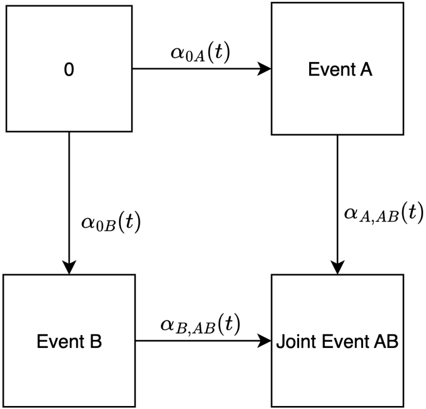

Figure: Interaction between two life history events

Likelihood Construction

Let the multi-state processes $X_i(t)$ be such as those described previously and observed over interval $[0, \tau_i]$ for individuals $i = 1, \ldots, n$.

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We can construct the likelihood using counting process notation.

Likelihood Construction

Let

$$0 < T_{bj}^{i1} < \cdots < T_{bj}^{iN_{bj}^i(\tau_i)} \leq \tau_i$$

be the times of $b \to j$ transitions, where $N_{bj}^i(t)$ is the number of direct transition $b \to j$ in $[0, t]$ for individual i.

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Let

$$Y_b^i(t) = I(X_i(t^-) = b)$$

and

$$Y_b(t) = \sum_{i=1}^{n} Y_b^i(t) = \#\text{ of individuals at risk in state } b \text{ at time } t^-$$

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Note: for $t > \tau_i$, we have $N_{bj}^i(t) = N_{bj}^i(\tau_i)$ and $Y_b^i(t) = 0$, so these quantities are defined on $(0, \infty)$.

Likelihood Construction

Denote $(\pi_b^i(0))$ to be the initial distribution for individual i.

Denote $f(Z_i)$ to be the density of time-fixed covariate Z_i.

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The likelihood is given by

$$\mathcal{L} = \prod_{i=1}^{n} \left\{ f(Z_i)\pi^i_{X_i(0)} \prod_{b \neq j} \prod_{k=1}^{N_{bj}^i(\tau_i)} \alpha_{bj}^i(T_{bj}^{ik}) \exp\left(-\int_0^{\tau_i} \alpha_{bj}^i(t) Y_b^i(t) dt \right) \right\} .$$

Andersen PK and Keiding N (2002)

Likelihood Construction

It is common to condition on the covariate Z_i and the initial value $X_i(0)$.

This way, we can omit $f(Z_i)\pi^i_{X_i(0)}$ in the likelihood:

$$\mathcal{L} \propto \prod_{i=1}^{n} \left\{ \prod_{b \neq j} \prod_{k=1}^{N^i_{bj}(\tau_i)} \alpha^i_{bj}(T^{ik}_{bj}) \exp\left(- \int_0^{\tau_i} \alpha^i_{bj}(t) Y^i_b(t) dt \right) \right\} .$$

Andersen PK and Keiding N (2002)

Likelihood: Left-Truncation and Right-Censoring

Incomplete observations:

- **Left-truncation**: individual i enters the study at some delayed time V_i
- **Right-censoring**: nothing is known about individual i after some time U_i

Likelihood: Left-Truncation and Right-Censoring

Incomplete observations:

- **Left-truncation**: individual i enters the study at some delayed time V_i
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To deal with left-truncation and right-censoring for this form of likelihood, we only need to modify the definition of the "at-risk indicator" $Y_b^i(t)$:

$$Y_b^i(t) = I(X_i(t^-) = b) \xrightarrow{\text{modify}} Y_b^i(t) = I(X_i(t^-) = b\ ,\ V_i \leq t \leq U_i)$$

Andersen PK and Keiding N (2002)

Model Specification

- **Markov Processes**: transition intensities only depend on the current state
  - constant and piecewise-constant transition intensities (common in econometrics, epidemiology, sociology and demography)
  - parametric transition intensities (less common, used in mortality studies in actuarial and some demographic studies)
  - nonparametric transition intensities (using Nelson-Aalen estimators)
  - regression models (e.g. Cox-type)
- **Semi-Markov Processes**: transition intensities depend on other time origins than $time = 0$, typically the time at entry to the present state

See Andersen PK and Keiding N (2002)

Application: PROVA Trial

- A clinical trial conducted to study the effect of propranolol and/or sclerotherapy versus no treatment on bleeding and survival in patients with liver cirrhosis.
- Between 1985 and 1989, 286 patients were randomized to different treatment combinations.
- An illness-death model is used, where the possible transitions are
  - no bleeding $\longrightarrow$ bleeding
  - no bleeding $\longrightarrow$ death
  - bleeding $\longrightarrow$ death

Source: PROVA Study Group. (1991). Propranolol and sclerotherapy for the prevention of recurrent variceal bleeding. New England Journal of Medicine, 324(22), 1532-1538.

Application: PROVA Trial

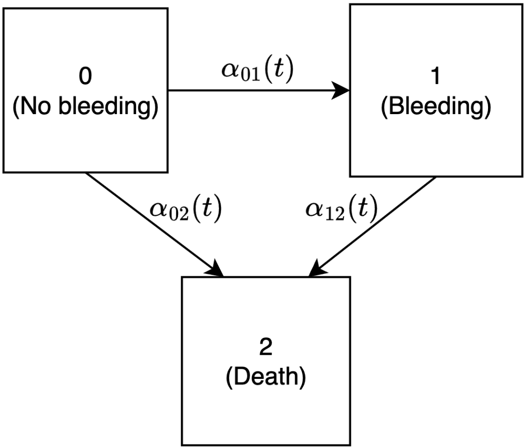

Figure: The illness-death model for PROVA trial

Application: PROVA Trial

Table 2 Estimates in Cox regression models for the PROVA data: bleeding intensity and intensity of death before bleeding

Covariate	$\alpha_{02}(t)$		$\alpha_{01}(t)$	
	$\hat{\beta}$	*SE*	$\hat{\beta}$	*SE*
Sclerotherapy	0.515	0.45	0.056	0.39
Propranolol	−0.319	0.57	−0.049	0.40
Both	0.908	0.42	−0.022	0.40
Sex(M = 1, F = 0)	0.928	0.42	0.154	0.32
Age(y)	0.0255	0.014	−0.0059	0.012

Source: Andersen PK and Keiding N (2002)

Remark: it is seen that the intensity of death without bleeding ($0 \to 2$) is affected by treatment

Application: PROVA Trial

Results for the $1 \to 2$ transition (death after bleeding):

Table 3 Estimates in Cox regression models for the PROVA data: intensity of death after bleeding, time variable is time since randomization

Covariate	$\hat{\beta}$	*SE*	$\hat{\beta}$	*SE*
Sclerotherapy	−0.736	0.66	−0.775	0.66
Propranolol	0.190	0.63	0.174	0.65
Both	0.711	0.54	0.427	0.58
Sex(M = 1, F = 0)	1.321	0.50	1.258	0.53
Age(y)	0.0398	0.018	0.0316	0.019
5 days > *WAIT*			2.844	0.68
10 days > *WAIT* ≥ 5 days			2.216	0.77

Table 4 Estimates in Cox regression models for the PROVA data: intensity of death after bleeding, time variable is waiting time in state 1

Covariate	$\hat{\beta}$	*SE*	$\hat{\beta}$	*SE*
Sclerotherapy	−0.810	0.61	−0.810	0.62
Propranolol	−0.400	0.59	−0.398	0.59
Both	0.613	0.49	0.691	0.51
Sex(M = 1, F = 0)	0.697	0.51	0.707	0.51
Age(y)	0.0030	0.016	0.0017	0.018
1 year > *TIME*			0.207	0.83
2 years > *TIME* ≥ 1 year			0.0459	0.79

Source: Andersen PK and Keiding N (2002)

Remark:

- Table 3 shows that Markovian assumption is violated: time-dependent covariates for the sojourn time (WAIT) spent in state 1 (bleeding) have a strong effect on the intensity
- Table 4 shows the results from a semi-Markov model: the mortality after bleeding is very high but decreases as time passed

Q & A

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References

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